

PES2UG24CS311

Neeraj R Rugi

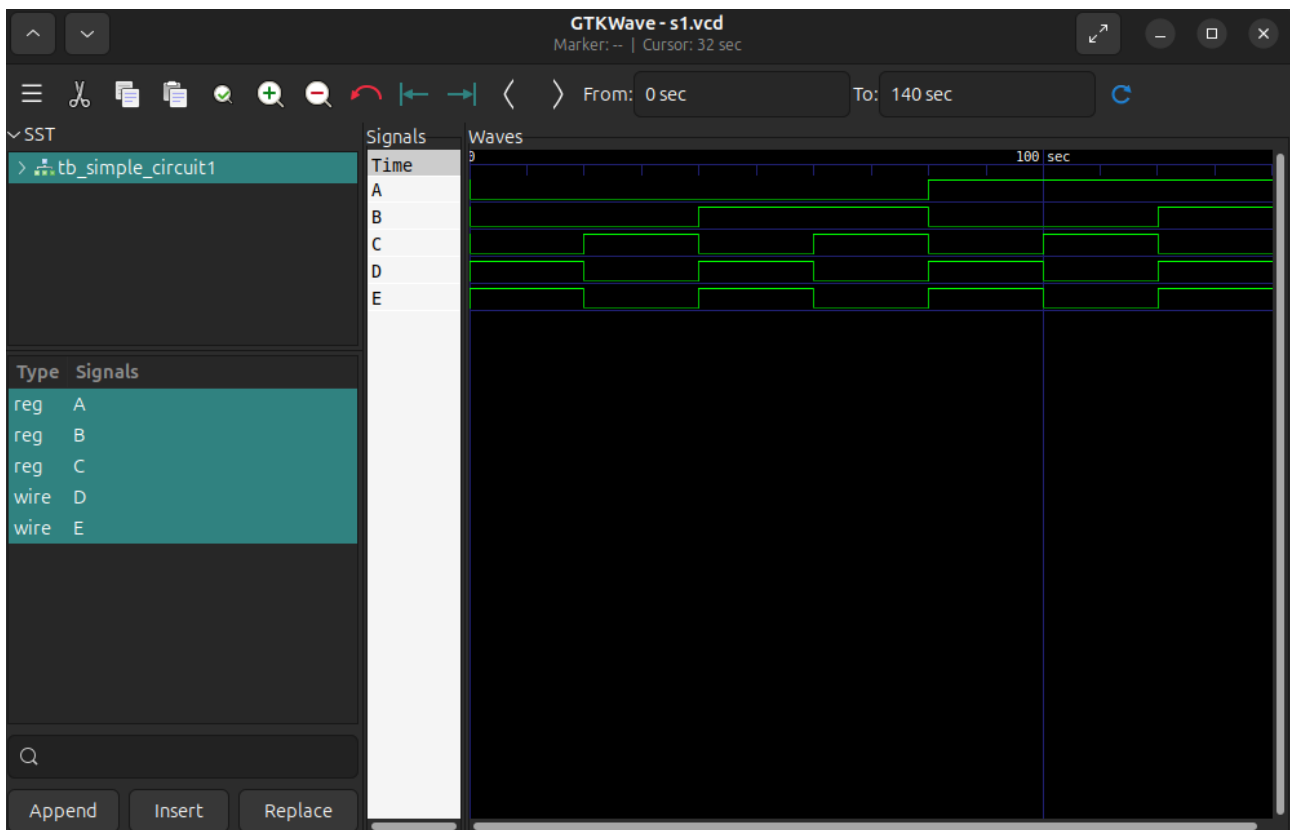
Lab – II

1. Simple-Circuit: 1

Terminal:

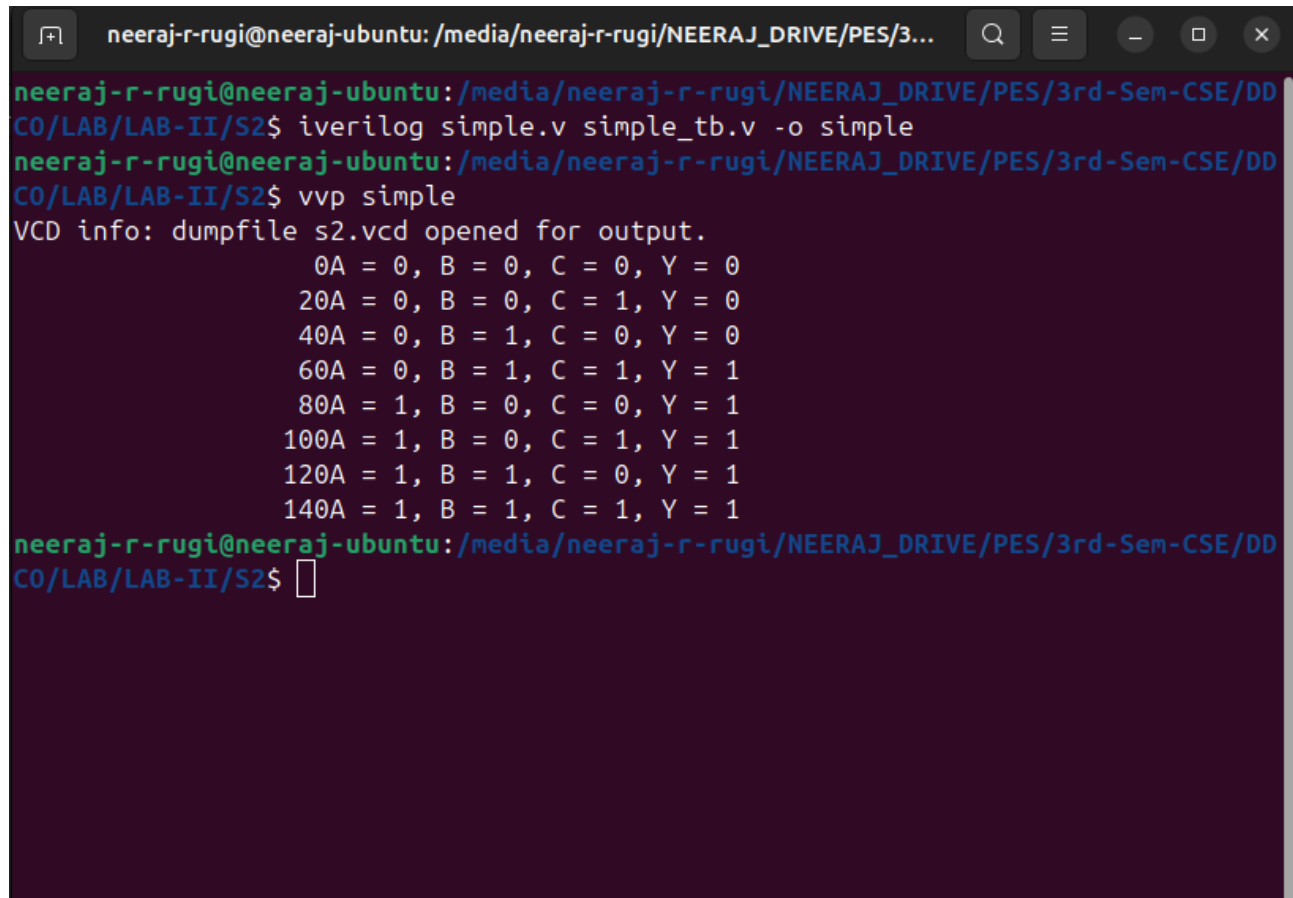
```
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S1$ iverilog simple.v simple_tb.v -o simple
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S1$ vvp simple
VCD info: dumpfile s1.vcd opened for output.
      0A = 0, B = 0, C = 0, D = 1, E = 1
     20A = 0, B = 0, C = 1, D = 0, E = 0
     40A = 0, B = 1, C = 0, D = 1, E = 1
     60A = 0, B = 1, C = 1, D = 0, E = 0
     80A = 1, B = 0, C = 0, D = 1, E = 1
    100A = 1, B = 0, C = 1, D = 0, E = 0
    120A = 1, B = 1, C = 0, D = 1, E = 1
    140A = 1, B = 1, C = 1, D = 1, E = 0
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S1$
```

GtkWave:



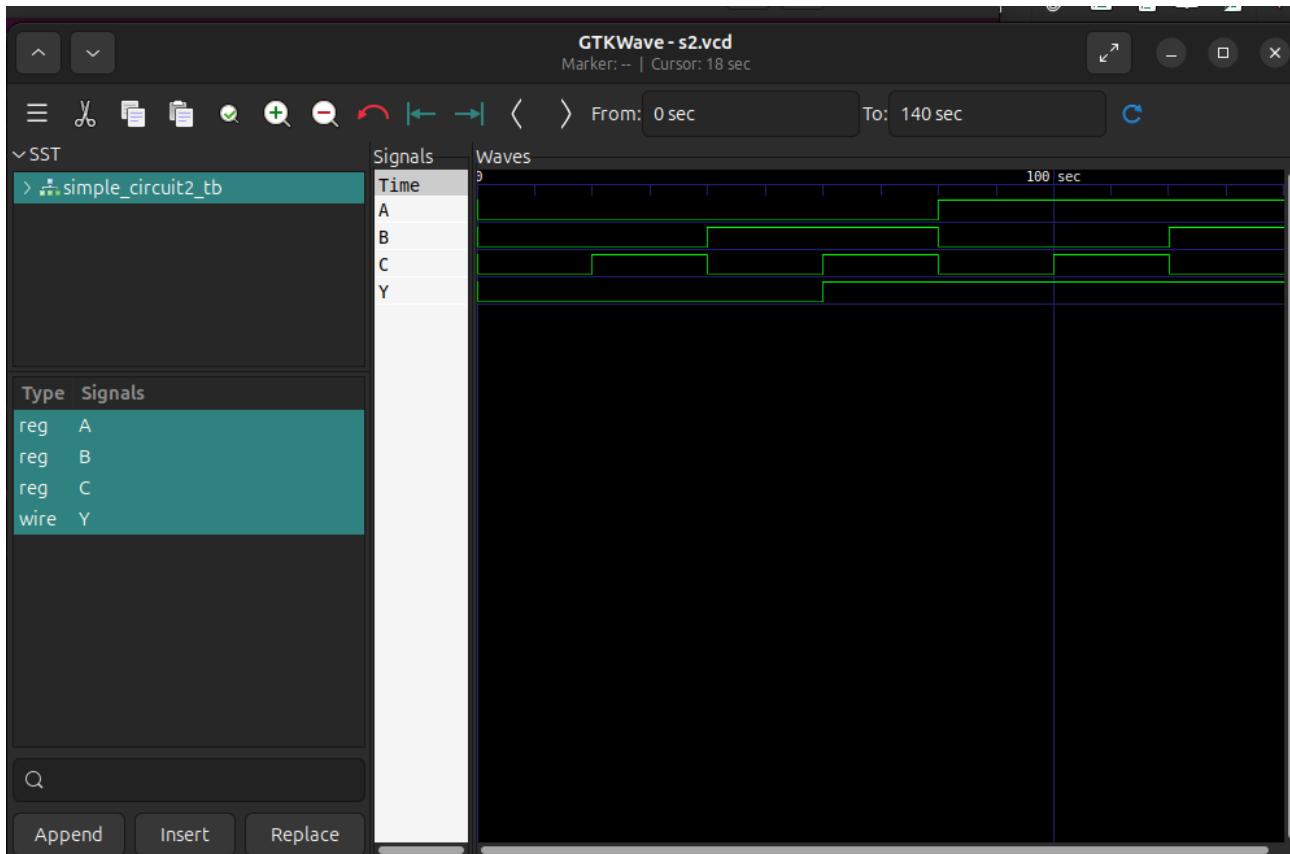
2. Simple-Circuit: 2

Terminal:



```
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3...
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S2$ iverilog simple.v simple_tb.v -o simple
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S2$ vvp simple
VCD info: dumpfile s2.vcd opened for output.
      0A = 0, B = 0, C = 0, Y = 0
     20A = 0, B = 0, C = 1, Y = 0
     40A = 0, B = 1, C = 0, Y = 0
     60A = 0, B = 1, C = 1, Y = 1
     80A = 1, B = 0, C = 0, Y = 1
    100A = 1, B = 0, C = 1, Y = 1
    120A = 1, B = 1, C = 0, Y = 1
    140A = 1, B = 1, C = 1, Y = 1
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S2$
```

GtkWave:



3.Assignment:

Code:

gate-Description:

```
module or2(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = a | b;  
  
endmodule  
module and2(c, a, b);  
    input a;  
    input b;  
    output c;  
    assign c = a&b;  
  
endmodule  
  
module fan(D, T, P, 0);  
input T, P, 0;  
output D;  
wire and_out;  
and2 G1(and_out, T, P);  
or2 G2(D, 0, and_out);  
  
endmodule
```

test-bench:

```
module simple_circuit3_tb;
    reg T, P, O;
    fan M1(D, T, P, O);
    initial begin
        #0 O = 1'b0;T = 1'b0;P = 1'b0;
        #20 O = 1'b0;T = 1'b0;P = 1'b1;
        #20 O = 1'b0;T = 1'b1;P = 1'b0;
        #20 O = 1'b0;T = 1'b1;P = 1'b1;
        #20 O = 1'b1;T = 1'b0;P = 1'b0;
        #20 O = 1'b1;T = 1'b0;P = 1'b1;
        #20 O = 1'b1;T = 1'b1;P = 1'b0;
        #20 O = 1'b1;T = 1'b1;P = 1'b1;
    end
    initial begin
        $monitor($time, " O = %b, T = %b, P = %b, D = %b", O, T, P, D);
    end
    initial begin
        $dumpfile("s3.vcd");
        $dumpvars(0, simple_circuit3_tb);
    end
endmodule
```

Terminal:

```
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S3$ iverilog simple.v simple_tb.v -o simple
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S3$ vvp simple
VCD info: dumpfile s3.vcd opened for output.
      0 0 = 0, T = 0, P = 0, D = 0
     20 0 = 0, T = 0, P = 1, D = 0
     40 0 = 0, T = 1, P = 0, D = 0
     60 0 = 0, T = 1, P = 1, D = 1
     80 0 = 1, T = 0, P = 0, D = 1
    100 0 = 1, T = 0, P = 1, D = 1
    120 0 = 1, T = 1, P = 0, D = 1
    140 0 = 1, T = 1, P = 1, D = 1
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-II/S3$
```

GtkWave:

